Week 10 Expanded Technical Summary and Slide Notes

Slide 1 Notes (Before Architecture) - Problem Statement

The bank is facing two linked risks at enterprise scale: payment fraud and anti-money laundering exposure. Transaction volume is massive, channels are distributed, and threats evolve quickly, so traditional batch controls are no longer sufficient for timely prevention.

Current-state pain points are operational and financial. Delayed detection creates avoidable losses, high false-positive rates overwhelm investigators, and legitimate customers experience unnecessary friction. At the same time, regulatory expectations require explainable alerts, complete audit trails, and repeatable controls.

The business objective is to move from reactive investigation to proactive prevention. The target state is a real-time decision system that can score risk, block high-risk events when necessary, and still preserve customer trust through consistent, transparent controls.

Slide 2 Technical Notes - Composite Architecture Roles

NiFi is the ingestion and flow-control layer. It connects to upstream banking systems, standardizes payloads, enriches events with metadata, and routes transactions to the correct downstream topics. Provenance tracking in NiFi also supports governance and troubleshooting when data quality issues appear.

Kafka is the streaming backbone. It holds transaction events in durable partitioned logs so multiple consumers can process the same stream independently. This decouples ingestion from analytics and enables real-time processing plus replay for backtesting and model updates.

Spark is the real-time computation engine. Structured Streaming consumes Kafka events, builds behavioral features, and applies rule logic plus machine-learning scoring. Spark windowing and state management are critical for identifying fraud patterns such as velocity spikes, geo anomalies, and unusual merchant behavior.

HBase is the low-latency operational store used during scoring. It maintains customer and account-level risk context, recent event summaries, and flags that must be retrieved in milliseconds. This allows each transaction to be scored against current behavior, not just static thresholds.

HDFS is the long-term distributed data lake. It stores raw and curated historical transactions at scale with fault tolerance. These records support retrospective analysis, model retraining, audit reconstruction, and regulatory evidence requirements.

Hive provides the SQL analytics and compliance reporting layer on top of HDFS. It is used for AML-oriented aggregations, periodic regulatory outputs, and investigator support datasets. Hive makes large-scale historical analysis accessible to analysts while keeping logic versionable and repeatable.

YARN is the cluster resource manager coordinating Spark and Hive workloads. It controls queue allocation, container scheduling, and workload isolation so latency-sensitive fraud pipelines remain stable even when heavy batch jobs are running.

Solr is the investigator search and triage interface layer. It indexes transaction and case artifacts for fast faceted search, entity linking, and retrieval. This shortens time-to-decision for analysts and converts model outputs into actionable case workflows.

As a composite machine, the platform works as an end-to-end pipeline: NiFi and Kafka move clean events, Spark and HBase execute live risk decisions, HDFS and Hive preserve analytical and compliance depth, YARN protects runtime performance, and Solr accelerates human investigation.

Slide 3 Notes (After Architecture) - Benefits & Conclusion

From a business perspective, the primary gain is speed-to-prevention. Risk scoring and policy actions occur within seconds, reducing loss exposure and helping teams intervene before fraudulent activity cascades across accounts or channels.

From an operational perspective, precision improves because model-driven and context-aware decisions reduce false positives. Investigation teams spend less time on noise and more time on high-value cases, which improves throughput and response quality.

From a governance perspective, the architecture is compliance-ready by design. Historical lineage in HDFS, structured reporting in Hive, and traceable event movement through the pipeline provide the evidence needed for AML and fraud-control audits.

Conclusion: this architecture is practical and scalable for a global bank because it combines real-time decisioning, resilient distributed infrastructure, and audit-grade analytics into one coherent operating model.